

HDOC DAY – 15

Algorithm:

1. Start.
2. Input a positive integer n.
3. Display the menu:
 - 1. Check whether all digits are even
 - 2. Check whether all digits are prime
4. Input the user's choice.
5. Use a switch statement based on the choice.
6. If choice = 1:
 - Examine each digit of the number.
 - Extract a digit using $\text{digit} = \text{temp} \% 10$.
 - If $\text{digit} \% 2 \neq 0$, then all digits are not even.
 - Remove the last digit using $\text{temp} = \text{temp} / 10$.
 - If no odd digit is found, all digits are even.
7. If choice = 2:
 - Examine each digit of the number.
 - Extract a digit using $\text{digit} = \text{temp} \% 10$.
 - Check whether the digit is one of 2, 3, 5, 7.
 - If any digit is not 2, 3, 5, 7, then all digits are not prime.
 - Remove the last digit using $\text{temp} = \text{temp} / 10$.
 - If every digit is 2, 3, 5, 7, all digits are prime.
8. For any other choice, display "Invalid choice".

9. Stop.

C program:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, temp, digit, choice;
```

```
    int flag = 1;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &n);
```

```
    /* Check whether the entered number is positive */
```

```
    if (n <= 0)
```

```
    {
```

```
        printf("Please enter a positive integer.\n");
```

```
        return 0;
```

```
    }
```

```
    printf("\nMENU\n");
```

```
    printf("1. Check whether all digits are even\n");
```

```
    printf("2. Check whether all digits are prime\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &choice);
```

```
temp = n;
```

```
switch (choice)
```

```
{
```

```
    case 1:
```

```
        /* Check every digit for evenness */
```

```
        while (temp > 0)
```

```
        {
```

```
            digit = temp % 10;
```

```
            if (digit % 2 != 0)
```

```
            {
```

```
                flag = 0;
```

```
                break;
```

```
            }
```

```
            temp = temp / 10;
```

```
        }
```

```
        if (flag == 1)
```

```
            printf("All digits are even\n");
```

```
        else
```

```
            printf("All digits are not even\n");
```

```
        break;
```

case 2:

```
/* Check whether every digit is prime */
```

```
while (temp > 0)
```

```
{
```

```
    digit = temp % 10;
```

```
    if (digit != 2 && digit != 3 &&
```

```
        digit != 5 && digit != 7)
```

```
    {
```

```
        flag = 0;
```

```
        break;
```

```
    }
```

```
    temp = temp / 10;
```

```
}
```

```
if (flag == 1)
```

```
    printf("All digits are prime\n");
```

```
else
```

```
    printf("All digits are not prime\n");
```

```
break;
```

default:

```
printf("Invalid choice\n");
```

```
}
```

```
return 0;
```

```
}
```

```
#include <stdio.h>

int main()
{
    int n, temp, digit, choice;
    int flag = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    /* Check whether the entered number is positive */
    if (n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    temp = n;

    switch (choice)
    {
        case 1:
            /* Check every digit for evenness */
            while (temp > 0)
            {
                digit = temp % 10;

                if (digit % 2 != 0)
                {
                    flag = 0;
                    break;
                }

                temp = temp / 10;
            }

            if (flag == 1)
                printf("All digits are even\n");
            else
                printf("All digits are not even\n");

            break;

        case 2:
            /* Check whether every digit is prime */
            while (temp > 0)
            {
                digit = temp % 10;

                if (digit != 2 && digit != 3 &&
                    digit != 5 && digit != 7)
                {
                    flag = 0;
                    break;
                }

                temp = temp / 10;
            }

            if (flag == 1)
                printf("All digits are prime\n");
            else
                printf("All digits are not prime\n");

            break;

        default:
```

Compilation and execution:

```
(kali㉿kali)-[~]  
$ vi h15.c  
  
(kali㉿kali)-[~]  
$ gcc h15.c -o h15  
  
(kali㉿kali)-[~]  
$ ./h15  
Enter a positive integer: 
```

Test case table:

Test case	Number	Choice	Expected output
1	12345	1	All digits are not even
2	24680	1	All digits are even
3	2357	2	All digits are prime
4	2379	2	All digits are not prime
5	2468	3	Invalid choice

```
(kali㉿kali)-[~]  
└─$ ./h15  
Enter a positive integer: 12345  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digits are prime  
Enter your choice: 1  
All digits are not even  
  
(kali㉿kali)-[~]  
└─$ ./h15  
Enter a positive integer: 24680  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digits are prime  
Enter your choice: 1  
All digits are even  
  
(kali㉿kali)-[~]  
└─$ ./h15  
Enter a positive integer: 2357  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digits are prime  
Enter your choice: 2  
All digits are prime  
  
(kali㉿kali)-[~]  
└─$ ./h15  
Enter a positive integer: 2379  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digits are prime  
Enter your choice: 2  
All digits are not prime
```